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Data Science Project Report

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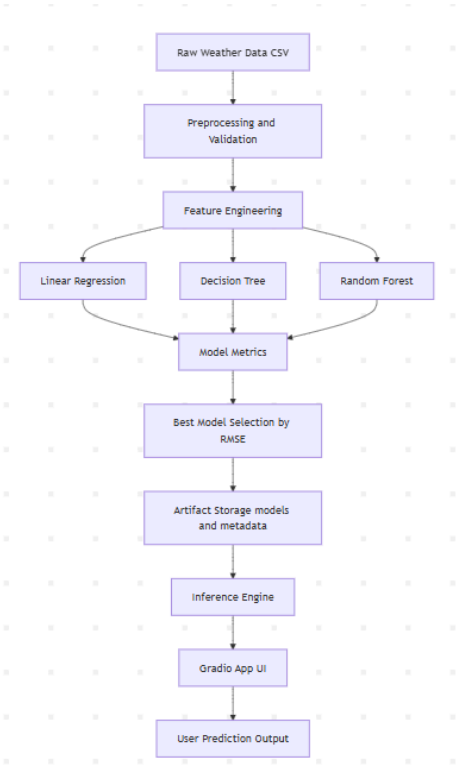
I. Introduction

Cambodia experiences clear seasonal and regional weather variation across provinces, and these changes directly influence agriculture, logistics, tourism, and daily planning. In this project, we apply a complete data science workflow to study historical weather behavior and build machine learning models for temperature forecasting.

The project focuses on predicting daily maximum temperature using weather and location signals such as minimum temperature, rainfall, wind speed, province, latitude, longitude, and calendar patterns. The final system is not only analytical but also practical: trained artifacts are deployed to an interactive interface so users can run predictions from structured inputs.

This work demonstrates how raw weather records can be transformed into useful decision support through preprocessing, feature engineering, model comparison, and lightweight deployment.

II. Project Flow Diagram



Project Flow Diagram

III. Problem Statement

Weather conditions are non-linear, time-dependent, and location-sensitive. A manual estimate of tomorrow's temperature is usually unreliable because multiple factors interact simultaneously, including rainfall, wind speed, local geography, and seasonal cycles.

The central challenge is to develop a forecasting model that learns useful patterns from historical Cambodian weather data while remaining practical to use in a real prediction workflow. The solution must balance:

- Predictive performance
- Stability across provinces and seasons
- Simplicity of deployment and usage
- Clear interpretability of model quality through metrics

IV. Objectives

The primary objectives of this project are to develop an accurate and reliable weather forecasting system for Cambodia, while applying data science techniques to analyze historical weather data and generate meaningful predictions:

- Build a clean and structured weather dataset for Cambodia.
- Engineer meaningful features from date, weather signals, and province location.
- Train and compare multiple regression models.
- Select the best-performing model using standard evaluation metrics.
- Deploy an easy-to-use interface for prediction and model selection.
- Document a reproducible pipeline from data loading to inference.

V. Methodology

5.1. Data Collection and Preparation

- Use the consolidated Cambodia weather dataset containing date, temperature, rain, wind speed, province, latitude, and longitude.
- Normalize column structure and parse date values.
- Clean data by handling invalid values and preserving realistic weather ranges.
- Apply basic quality constraints, including non-negative rain and wind speed, and valid geographic bounds.

5.2. Feature Engineering

- Create calendar features such as year, month, day, and day of week.
- Add cyclical encodings for month and day of week to represent seasonality.
- Encode province information for region-aware modeling.
- Build a model-ready numeric matrix with consistent feature columns for training and inference.

5.3. Feature Engineering

- Split prepared data into training and testing sets.
- Train candidate models and evaluate them with MAE, RMSE, and R2.
- Save model artifacts and metadata for reproducible inference.

5.4. Inference and UI

- Save trained models and metadata as artifacts.
- Load artifacts in a Gradio app for user-driven prediction.
- Allow model selection and configurable weather inputs for scenario-based prediction.

VI. Modeling

Three regression models are trained and compared:

6.1. Linear Regression

- Baseline model used to measure linear relationships between input variables and target temperature.
- Strength: simple and interpretable.
- Limitation: weaker performance for complex non-linear weather behavior.

6.2. Decision Tree Regressor

- Captures non-linear splits and interaction effects between weather signals.
- Strength: interpretable branching behavior and flexible fitting.
- Limitation: can overfit if not regularized.

6.3. Random Forest Regressor

- Ensemble of multiple trees to improve robustness and generalization.
- Strength: strong performance and reduced variance compared to a single tree.
- Limitation: less directly interpretable than a linear model.

Based on current project artifacts, Random Forest is selected as the best model by RMSE for temp_max prediction. Evaluation is conducted using:

- **MAE** to capture average absolute error magnitude
- **RMSE** to penalize larger prediction errors
- **R2** to measure explained variance

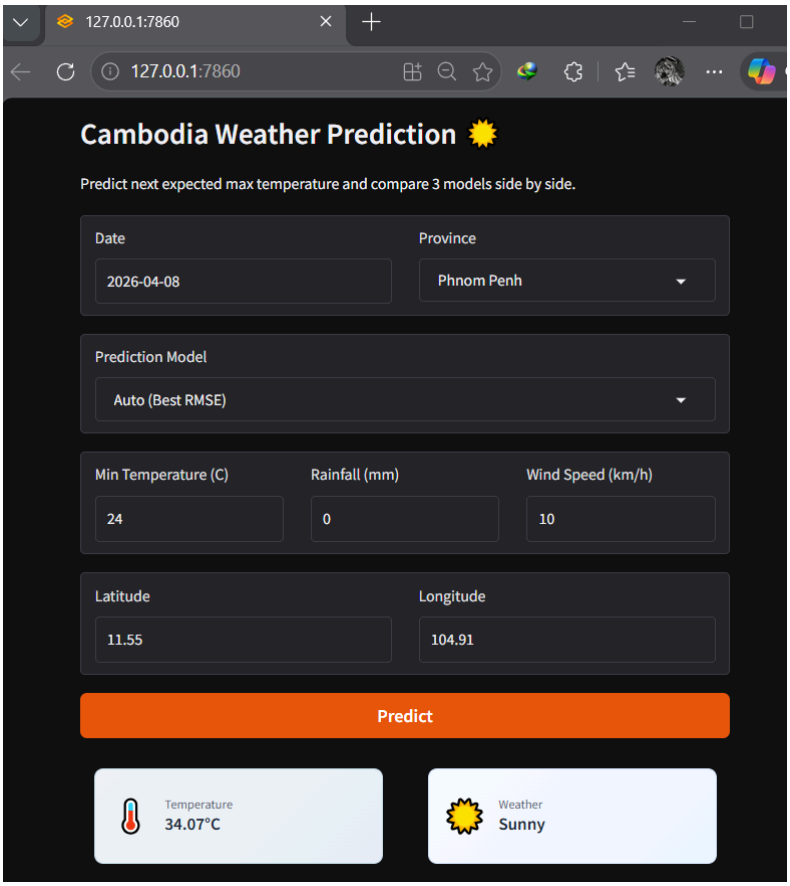
This multi-metric approach ensures model choice is not based on a single score and helps balance accuracy and reliability.

VII. Conclusion

This project delivers a full forecasting pipeline, from raw data transformation to an interactive prediction interface. It demonstrates that structured preprocessing, region-aware feature engineering, and model comparison can produce a practical weather prediction workflow for Cambodia.

The modeling results show that ensemble learning, specifically Random Forest, currently provides the best predictive quality among tested candidates for maximum temperature estimation. The project is therefore suitable as both a learning case study and a functional prototype for weather-informed planning.

For future improvement, the next priorities are stronger time-aware validation, richer climate features, and expanded automated testing to improve robustness and long-term maintainability.



(Gradio Prototype)

Resources

Api Source - https://open-meteo.com/

Github - https://github.com/obviouslynotdim/WeatherForecast-AnalysisDS

Gradio - https://www.gradio.app/